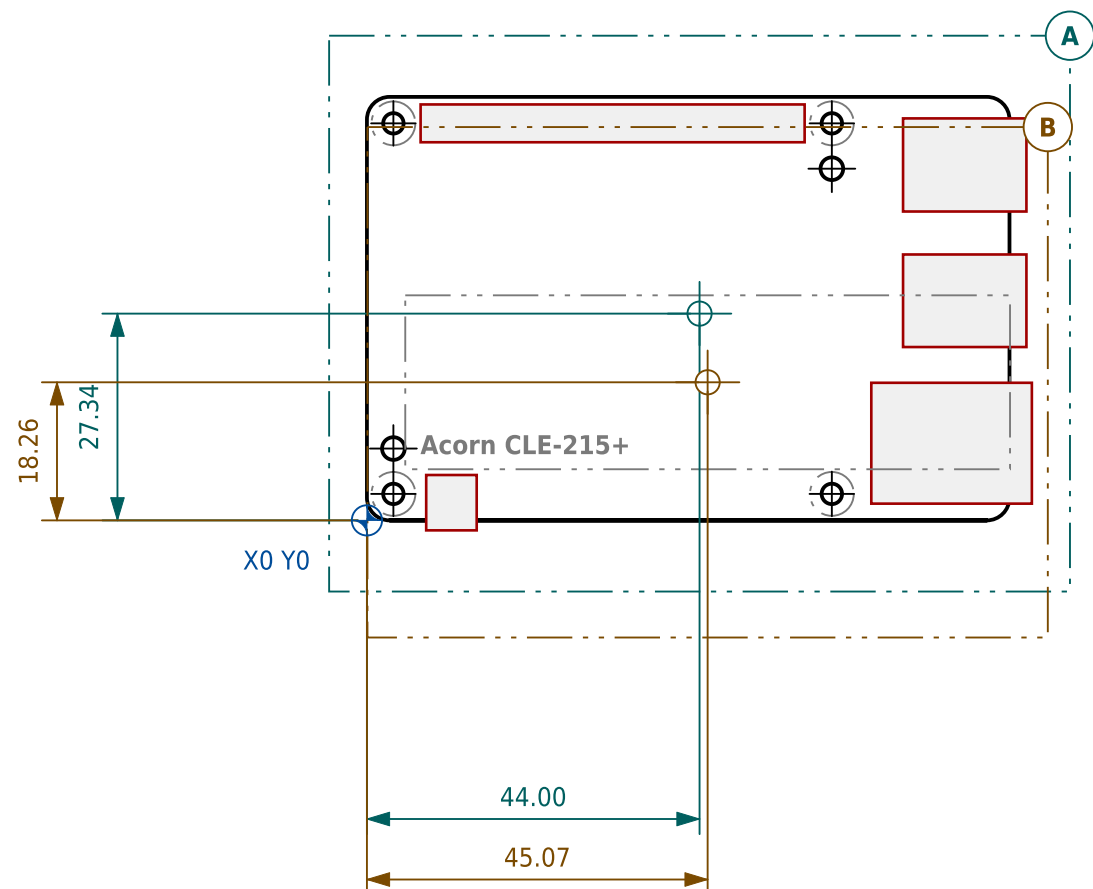




NOTES

1. The camera looks straight down. X and Y locate the optical axis in the subject's plan frame; Z is the lens height above the plane its target lies in. ASSUMED: Z is to the entrance pupil, which no vendor locates; set it from the lens face.
2. A frame is the smallest rectangle of the sensor's own 4:3 holding its target plus 5.00 mm all round; LONG says which way the camera is turned. It is the shape of the picture, not of the optics, so both lenses share it and differ only in Z.
3. PLANE, which Z is measured from: A/B from the ACORN CARD's top face, not the Pi's -- nobody publishes how far the card stands above the Pi, so measure the stack. Set from the Pi's face the camera sits too low, and the picture at the card loses the ends of it. A stand set h mm too low covers only $(Z - h) / Z$ of what is drawn, so the error always loses the edges.
4. FOCUS: Raspberry Pi give the depth of field as "Approx 1 m to infinity", so 1000 mm is the nearest this lens focuses. Every height here is nearer than that, so a stock OV5647 cannot focus on any of these boards. Use one whose lens focuses closer.
5. AUTOFOCUS: an autofocus OV5647 is a different part -- Arducam's B0176, declared 54 x 44 deg against the stock lens's 53.50 x 41.41, with a motorised lens and a close focus setting. Arducam publish no near limit and no lens height for it. Raspberry Pi's focusable modules, other sensors, are given as "Approx 10 cm to infinity" and "Approx 5 cm to infinity".

NOTES (continued)

6. The 120 deg lens's declared pair is not self-consistent: 120 across a 4:3 sensor implies 104.82 down it, not 90. Z is the greater of what the two ask for, so the picture runs OVER the rectangle -- A by 29.31 mm in X, B by 26.91 mm in X -- which is still covered.
7. The "65 deg" name is the stock lens's diagonal, which no vendor prints. DERIVED: $2 \times \arctan(\sqrt{3.76^2 + 2.74^2}) / 3.60 = 65.74$ deg from the image area, 64.42 from the pixel array.
8. Quotes here are transliterated to ASCII: x for the multiplication sign, um for micro, infinity for its symbol. The cached pages carry them as printed.
9. Frame A, Whole assembly: Pi 5 with a Waveshare PoE M.2 HAT+ (B) and the Acorn seated in it; the HAT is the Pi's own 85 x 56 mm and its 2280 standoff reaches 88.00.
10. Frame B, Acorn card: 23 x 80 mm: the M.2 specification's Type 2280 outline with SQRL's own extra millimetre of width.
11. The Acorn's own LED positions are not published: SQRL issued no mechanical drawing and their site is gone, so frame B is the card, not its indicators. Its seated position and the assembly's far edge are accessories/parts.py's, which ACC-HAT-M2POE is drawn from too.

SOURCES

1. Board geometry: raspberry_pi/boards.py - Pi 5 outline and connectors; see RPI-5.
2. PCI Express M.2 Specification: PCI-SIG, Revision 1.0, 1 November 2013 - Figure 13: Type 2280 is 22 x 80, both +/-0.15.

FRAMES, AND Z ABOVE THE FRAME PLANE, mm

	FRAME	RECTANGLE mm	LONG	AXIS X	AXIS Y	Z 65	Z 120
A	Whole assembly	98.00 x 73.50	X	44.00	27.34	97.2	36.8
B	Acorn card	90.00 x 67.50	X	45.07	18.26	89.3	33.8

LENSES

LENS	H deg	V deg	FOCUS	NEAR mm
65 deg (stock lens)	53.50	41.41	Fixed	1000
120 deg (wide lens)	120.00	90.00	Fixed	not published
Autofocus	54.00	44.00	Auto	not published

LEGEND

- Subject outline
- LED, display or connector
- Adjacent part or connector body
- - - Frame A, the rectangle the picture must cover
- - - Frame B, the rectangle the picture must cover
- Dimension, extension and leader

SOURCES (continued)

3. SQRL Acorn CLE-215+ product page:
<https://web.archive.org/web/2020/http://www.squirrelsresearch.com/acorn-cle-215-plus/> - Captured 2020; the site is gone. Quoted: "it is one millimeter wider than the official specifications."
4. Waveshare PoE M.2 HAT+ (B) dimension drawing:
<https://www.waveshare.com/w/upload/d/d9/PoE-M.2-HAT-Plus-B-details-size.jpg> - Annotated 85.00, 56.00 and 3.00, "Unit: mm": the 3.00 is the standoff past the board edge. It dimensions no height.
5. Raspberry Pi camera documentation:
<https://web.archive.org/web/20241230011811/https://www.raspberrypi.com/documentation/accessories/camera.html> - Camera Module 1 column: the "OmniVision OV5647", its "2592 x 1944 pixels" at "1.4 um x 1.4 um", "3.76 x 2.74 mm", "3.60 mm +/- 0.01", "53.50 +/- 0.13 degrees", "41.41 +/- 0.11 degrees", focus "Fixed", "Approx 1 m to infinity".
6. Arducam 5MP OV5647 documentation:
<https://docs.arducam.com/Raspberry-Pi-Camera/Native-camera/5MP-OV5647/> - Product catalogue: B0033 "Stock Lens 54 (H) x 41 (V) Fixed Focus", B006604 "120 (H) x 90 (V)", B0176 "54(H)x44 (V) Auto Focus".
7. Arducam motorized focus camera:
<https://docs.arducam.com/Raspberry-Pi-Camera/Motorized-Focus-Camera/Motorized-Focus-Camera/> - Quoted: "you can understand it the same as autofocus"; its OV5647 guide adds a voice-coil device tree line and a close-focus range.

CAMERA OVER THE ACORN CLE-215+

Camera Module OV5647, 65 and 120 degree lenses

DRAWN generated		VERSION v0.0-326-gcbe28df		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO RPICAM-OVER-ACORN				REV A
SUBJECT Acorn CLE-215+, Pi 5			GENERAL TOLERANCE Pi 5 +/-0.20, card +/-0.20 DERIVED, Z DERIVED				